Traditional Knowledge Systems of India

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It is now recognized that western criteria are not the sole benchmark by which other cultural knowledge should be evaluated. While the term ‘traditional’ sometimes carries the connotation of ‘pre-modern’ in the sense of ‘primitive’ or ‘outdated’, many of the traditional sciences and technologies were in fact quite advanced even by western standards as well as better adapted to unique local conditions and needs than their later ‘modern’ substitutes. ***In countries with ancient cultural traditions, the folk and elite science were taken as part of the same unified legacy, without any hegemonic categorizations.***

However, modernization has homogenized various solutions, and this loss of ideas is similar to the destruction of biodiversity. Colonizers systematically derogated, exterminated or undermined the local traditional science, technology and crafts of the lands and people they plundered, because of their intellectual arrogance, and also to control and appropriate the economic means of production and the social means of organization.

Modern societies created hegemonic categories of science verses magic, technology verses superstitions etc., which were arbitrary and contrived. But many anthropologists who have recently worked with so-called ‘primitive’ peoples have been surprised to learn of some of their highly evolved and sophisticated technologies. The term ‘Traditional Knowledge System’ was thus coined by anthropologists as a scientific system which has its own validity, in contradistinction to ‘modern’ science.

The United Nations University proposal defines ‘Traditional Knowledge Systems’ as follows:

“Traditional knowledge or ‘local knowledge’ is a record of human achievement in comprehending the complexities of life and survival in often unfriendly environments. Traditional knowledge, which may be technical, social, organizational, or cultural was obtained as part of the great human experiment of survival and development.”

Laura Nader describes the purpose of studying Traditional Knowledge Systems (TKS): “The point is to open up people’s minds to other ways of looking and questioning, to change attitudes about knowledge, to reframe the organization of science — to formulate a way of thinking globally about traditions.”

Historical Background

Modern science can perhaps be dated to Newton’s times. But Traditional Knowledge Systems date from more than 2 million years, when Homo habilis started making his tools and interacting with nature. Since the dawn of history, different peoples have contributed to different branches of science and technology, often in a manner involving interactive contacts across cultures separated by large distances. This interactive influence is becoming clearer as the vast extent of global trade and cultural migration across large distances is being properly recognized by researchers.

However, one finds that generally the history of science as commonly taught is mostly Eurocentric. It typically consists of two phases: It starts with Greece, neglecting the influences of others upon

Greece. Then it ‘fast forwards’ many centuries to the Enlightenment period around 1500, to claim modern science as an exclusively European triumph, by neglecting the influence of others, especially India, upon the European Renaissance and Enlightenment. The European Dark Ages is presumed to be dark worldwide, when in fact, the rest of the world thrived with innovation and prosperity while Europe was at the peripheries until the conquest of America in 1492.

Thanks to especially the work of Joseph Needham, China’s contributions to global knowledge have recently become known to a wide range of scholars. Even more recently, thanks largely to Arab scholars, the important role of Islamic empires in the transmission of ideas into Europe has become better appreciated. However, in the latter case, many discoveries and innovations of India, as acknowledged by the Arab translators themselves, are often depicted as being of Arab origin, when in fact, the Arabs often retransmitted what they had learnt from India over to Europe.

Therefore, the vast and significant contributions made by the Indian sub-continent have been widely ignored. The British colonizers could never accept the fact that Indians were highly civilized even in the third millennium BC, when the British were still in a barbarian stage. Such acknowledgment would destroy the civilizing mission of Europe that was the intellectual premise for colonialization.

British Indologists did not study TKS, except to quietly document them as systems competing with their own, and to facilitate the transfer of technology into Britain’s industrial revolution. What was found valuable was quickly appropriated (see examples below), and its Indian manufacturers were forced out of business, and this was in many instances justified as civilizing them. Meanwhile, a new history of India was fabricated to ensure that present and future generations of mentally colonized people would believe in the inherent inferiority of their own traditional knowledge and in the superiority of the colonizers’ ‘modern’ knowledge. This has been called Macaulayism, named after Lord Macaulay who successfully championed this strategy of Britain most emphatically starting in the 1830s.

Because it became difficult for Europeans to ignore the massive archaeological evidence of classical Indian science and technology, they propounded that the Indus civilization had to be a transplant from the Egyptian and Mesopotamian civilizations. These constructions in historiography have tended to be cumulative rather than re-constructive, i.e. more layers were constructed without re-examining or correcting prior ones. Unfortunately, since Independence there has not been much improvement in such distortions of history, and this has continued to negatively impact the understanding and appreciation of TKS. Many in India’s intellectual elite continue to promote the notion that pre-colonial India was feudalistic, pre-rational, and by implication in need of being invaded for its own benefit.

This has created a climate in which entrenched prejudice against TKS still persists in contemporary society. For example, according to TKS activist Madhu Kishwar, India’s government today continues to make many TKS illegal or impossible to practice. Even after independence, many British laws against TKS have continued, even though their original intent was to destroy India’s massive domestic industry and foreign trade and to replace them with Britain’s industrial revolution. It is significant to note that today less than 10% of India’s labor works in the ‘organized sector’, namely as employees of a company. The remaining 90% are individual freelancers, contract laborers, private entrepreneurs, and so on, many of whom still practice their traditional trades.

However, given the perpetuation of colonial laws that render much of their work illegal, they are highly vulnerable to all sorts of exploitation, corruption, and abuse. The descendents of India's traditional knowledge workers, who built massive cities, technologies, and dominated world trade for centuries, are today de-legitimized in their own country under a democratic government. ***Many of today's poor jatis, such as textile, masonry, and metal workers, were at one time the guilds that supplied the world with so many and varied industrial items.***

It is important to note that amongst all the conquered and colonized civilizations of the Old World, India is unique in the following respect: Its wealth was industrial and created by its workers' ingenuity and labor. In all other instances, such as the Native Americans, the plunder by the colonizers was mainly of land, gold and other natural assets. But in India's case, the colonizers had a windfall of extraordinary profit margins from control of India's exports, taxation of India's economic production, and eventually the transfer of technology and production to the colonizer's home. This comprised the immense transfer of wealth out of India.

From being the world's major exporting economy (along with China), India was reduced to an importer of goods; from being the source of much of the economic capital that funded Britain's industrial revolution, it became one of the biggest debtor nations; from its envied status as the wealthiest nation, it became a land synonymous with poverty; and from the nation with a large number of prestigious centers of higher education that attracted the cream of foreign students from Eurasia, it became the land with the highest number of illiterate persons. This remains a major untold story. The education system's subversion of India's TKS in its history and social studies curricula is a major factor for the stereotyping about India. Even when told of these things, few westerners and elitist Indians are willing to believe them, as the prejudices about India are too deeply entrenched.

The Global Problem Today

The present day globalizing economy with its mass media glorification of the western lifestyle is resulting in the homogenization of human 'wants' and in unachievable expectations. Conventional western technology by itself cannot deliver or sustain this false promise to the world, for several reasons:

Westernized living is unachievable by billions of poor humans, because the capital required simply does not exist in the world, and the trickle down effect is too slow to reach the bottom tier where most of humanity lives.

Western civilization depends upon inequality — there must be cheap labor 'somewhere else', and cheap natural resources purchasable from somewhere, without regard to the big picture of world society or global ecology. This practical necessity of the present-day global capitalist system conflicts with the equal rights of states and persons long theorized and promoted. All sorts of reasons are offered against such drastic proposals as opening all borders and allowing free competition among all available laborers, contradicting the 'freedom' position so popular in theory.

The western economic development model demands 'growth' to sustain valuations in the stock markets, and growth cannot be indefinite. A steady state economy in zero growth equilibrium would devastate the wealth of the west, since the financial models are predicated on growth.

Even if the above obstacles could be overcome and the world's six billion persons were to achieve western lifestyle, it would be unsustainable for the planet's natural resources to sustain.

When Gandhi was asked whether he would like India to develop a lifestyle similar to England's, his reply may be paraphrased as follows: The British had to plunder the Earth to achieve their lifestyle. Given India's much larger population, it would require the plunder of many planets to achieve the same.

If the idealized lifestyle is unavailable to all humanity, then on what basis (morally, intellectually, and in terms of practical enforcement) do a few countries hope to sustain their superiority over others so as to maintain such a lifestyle? The point is that employing TKS is an imperative for humanity at large, while reducing global dependence on inequitable and resource draining 'advanced' knowledge systems.

We have to study, preserve, and revive the Traditional Knowledge Systems for the economic betterment of the world in a holistic manner, as these technologies are eco-friendly and allow sustainable growth without harming the environment. India's scientific heritage needs to be brought to the attention of the educated world, so that we can replace the Eurocentric history of science and technology with an honest globalization of ideas. This goal requires generations of new research in these fields, compilation of existing data, and dissemination through books, seminars, websites, articles, films, etc.

Indian Contributions to Global Science

Civil Engineering: The Indus-Sarasvati Civilization was the world's first to build planned towns, with underground drainage, civil sanitation, hydraulic engineering, and air-cooling architecture. Oven baked bricks were invented in India in approximately 4,000 BC. From complex Harappan towns to Delhi's Qutub Minar and other large projects, India's indigenous technologies were very sophisticated in design, planning, water supply, traffic flow, natural air conditioning, complex stonework, and construction engineering.

Metal Technologies: They pioneered many tools for construction, including the needle with hole at the pointed end, hollow drill, and true saw. Many of these important tools were subsequently used in the rest of the world, centuries later during Roman times. India was the first to produce rust-free iron. In the mid-first millennium BC, the Indian wootz steel was very popular in the Persian courts for making swords. The British sent teams to India to analyze the metallurgical processes that were later appropriated by Britain. Making India's metal works illegal was motivated partly by the goal to industrialize Britain, but also because of the risk of gun manufacturing by potential nationalists. India's exporting steel industry was systematically dismantled and relocated to Britain.

Textiles: India's textile exports were legendary. Roman archives contain official complaints about massive cash drainage because of imports of fine Indian textiles. One of the earliest industries relocated from India to Britain was in textiles, and it became the first major success of the Industrial Revolution, with Britain replacing India as the world's leading textile exporter. Many of the machines built by Britain used Indian designs that had been improved over long periods. Meanwhile, India's textile manufacturer's were de-licensed, even tortured in some cases, over-taxed, regulated, etc., to 'civilize' them into virtual extinction.

Shipping and Ship Building: India participated in the earliest known ocean based trading systems. Regarding more recent centuries, it is known to scholars but not to the general public that Vasco da Gama's ships were captained by a Gujarati sailor, and much of Europe's 'discovery' of navigation was in fact an appropriation of pre-existing navigation in the Indian Ocean, that had been a thriving trade system for centuries before Europeans 'discovered' it. Some of the world's largest and most sophisticated ships were built in India and China. The compass and other navigation tools were already in use at the time. ('Nav' is the Sanskrit word for boat, and is the root word in 'navigation', and in 'navy', although etymology is not a reliable proof of origin.)

Water Harvesting Systems: Scientists estimate that there were 1.3 million man-made water lakes and ponds across India, some as large as 250 square miles. These are now being rediscovered using satellite imagery. These enabled most of the rain water to be harvested and used for irrigation, drinking, etc. till the following year's rainfall. Village organizations managed these resources, but this decentralized management was dismantled during the colonial period, when tax collection, cash expropriation, and legal enforcements became the primary function of the new governance appointed by the British. Recently, thousands of these 'talabs' have been restored, and this has resulted in a re-emergence of abundant water year round in many places. (This is a very different approach compared to the massive modern dams built in the name of progress, that have devastated the lives of millions.)

Forest Management: Many interesting findings have recently come out about the way forests and trees were managed by each village and a careful method applied to harvest medicines, firewood, and building material in accordance with natural renewal rates. There is now a database being built of these 'sacred groves' across India. Again, it's a story of an economic asset falling into disuse and abuse because of dismantling the local governance and uprooting respect for traditional systems in general. Massive logging by the British to export India's timber to fund the two world wars and other civilizing programs of the empire are never mentioned when scholars try to explain India's current ecological disasters. The local populations had been quite sophisticated in managing their ecology until they were dis-empowered.

Farming Techniques: India's agricultural production was historically large and sustained a huge population compared to other parts of the world. Surpluses were stored for use in a drought year. But the British turned this industry into a cash cow, exporting massive amounts of harvests even during shortages, so as to maximize the cash expropriation. This caused tens of millions to die of starvation while at the same time India's food production was exported at unprecedented rates to generate cash. Also, traditional non-chemical based pesticides have been recently revived in India with excellent results, replacing Union Carbide's products in certain markets.

Traditional Medicine: This is now a well-known and respected field. Much re-legitimizing of Indian medicine has already been done, thanks to many western labs and scientists. Many multinationals no longer denigrate traditional medicine and have in fact been trying to secure patents on Indian medicine without acknowledging the source.

Mathematics, Logic and Linguistics: Besides other sciences, Indians developed advanced math, including the concept of zero, the base-ten decimal system now in use worldwide, and many important trigonometry and algebra formulae. They made several astronomical discoveries. Diverse schools of logic and philosophy proliferated. India's Panini is acknowledged as the founder of linguistics, and his Sanskrit grammar is still the most complete and sophisticated of any language in the world.

There were numerous other indigenous Indian industries. India's manufactured goods were highly prized around the world. We must evaluate the historical importance of these TKS based on their economic value for their time, when their importance could be compared to today's high tech industry. India's own English educated elite should be made aware of this to shed their Macaulayite inferiority complexes. Furthermore, the development, refinement and extension of TKS offer potential benefits capable of resolving or diminishing some of the inequities in modern societies worldwide.

Folk Science

Besides the above examples of Indian contributions to the very foundations of so-called 'western' science, another category of Traditional Knowledge Systems is non-literate folk science. Western science as a whole has condemned and ignored anything that it did not either appropriate or develop, as being magic and superstition. However, in countries such as India that have cultural continuity, ancient traditions survive with a rich legacy of folk science.

In North America and Australia, where original populations have been more than decimated, such continuity of folk tradition was disrupted. In Western nations with large colonies in the Old and New Worlds, such knowledge systems were looked down upon. It is this prejudice that subverts the importance of folk science, and ridicules it as superstition. The process of contrasting western science with folk knowledge systems extends to the demarcation of knowledge systems in different categories of science versus religion, rational versus magical, and so on. But we need to insist that these western imposed hegemonic categories are contrived and artificial.

Western science seldom realized that non-literate folk science preserves the wisdom gained through millennia of experience, direct observation, and has been transmitted by word of mouth. Development projects based solely on new technologies are pushing the Traditional Knowledge Systems towards extinction. This traditional wisdom of humankind needs to be preserved and used for our survival.

Westernized 'experts' go to non-literate cultures assuming them to be 'knowledge blanks' which need to be programmed with modern science and technology. Ramkrishnan, the renowned ecologist, humbly admitted that the ecological management practiced today by the tribes of the northeastern states of India is far superior to anything he could teach them. A good example in this regard is the alder (Alnus nepalensis), which has been cultivated in the jhum (shifting cultivation) fields by the Khonoma farmers in Nagaland for centuries. It has multiple usages for the farmers, since it is a nitrogen-fixing tree and helps to retain the soil fertility. Its leaves are used as fodder and fertilizer, and it is also utilized as timber. One could cite numerous such examples. Unfortunately, many plants which the tribes traditionally cultivated for specific benefits have now disappeared in the name of progress.

The vast majority of modern medicines patented by western pharmaceutical firms are based on tropical plants. The most common method to select candidates for detailed testing has been for western firms to scout tropical societies, seek out established 'folk' remedies, and to subject these to 'western scientific legitimizing'. In many cases, patents owned by multinationals are largely for isolating the active ingredients in a lab, and going through rigorous protocols of testing and patent filing.

While this is an important and expensive task, and deserves credit, these are seldom independent discoveries from scratch. Never has the society that has truly discovered it through centuries of empirical testing and trial and error received any recognition, much less any share of royalty. India's recent fights in international courts, over western patents of its traditional intellectual property in agriculture and medicine, have brought much needed publicity for this arena.

Colin Scott writes: "With the upsurge of multidisciplinary interest in 'traditional ecological knowledge', models explicitly held by indigenous people in areas as diverse as forestry, fisheries, and physical geography are being paid increasing attention by western science specialists, who have in some cases established extremely productive long-term dialogues with local experts. The idea that local experts are often better informed than their western peers is at last receiving significant acknowledgment beyond the boundaries of anthropology."

But in too many cases, western scholars reduce India's experts to 'native informants' destined to live below the glass ceiling: the pandit as native informant to the western Sanskritist; the poor woman in Rajasthan as native informant to the western feminist seeking to cure her of her tradition; the herbal farmer as native informant to the western pharmaceutical firm appropriating medicines for patents; etc. Given their poverty in modern times, these 'native informants' dish out what the western scholar expects to hear in order to fit his/her model, because in return they receive gifts, rewards, compensation, recognition, and even trips and visas in many cases.

Rarely have western scholars acknowledged India's knowledge bearers as fellow scientists and equal partners, as co-authors or as co-panelists. This competitive obsession to make 'original' discoveries and to put one's name on publications has exacerbated the tendency to appropriate with one hand, while denigrating the source with other hand so as to hide the plagiarism. We have referred to this as 'academic arson'.

Rituals as Knowledge Transmitters

Villagers in remote areas like Uttaranchal have events called 'Jagars', in which the Jagaria sends the Dangaria into a sort of trance. The Dangaria then helps sort out problems, provides remedies for ailments, resolves social conflicts of the village society etc. One could dismiss this as superstition; but this is also considered a traditional method of reaching the unconscious. Does the Jagaria use his spiritual powers to reach and tap the unconscious region of the mind of the Dangaria? Or, as propounded by Vaclav Havel, did these rituals represent the attempts of ancient humans to come to terms with the unknown, the non-rational, and the unconscious parts of our beings? Were these devices useful to invoke lost memories of the ancient past?

We are, therefore, not willing to dismiss Jagar as some mumbo-jumbo, but a phenomenon worth scientific investigation. This should be an important scientific research connecting Traditional Knowledge Systems to Inner Sciences. Ironically, from Jung onwards, many westerners have studied and appropriated these traditional 'inner sciences', renamed and repackaged them. Meanwhile, the original discoverers and practitioners have been dismissed as primitive societies awaiting cure by westernization.

Myths and Legends:

Myths and legends sometimes represent the attempts of our ancestors to explain the scientific observations that they made about the world around them and transmitted to the future. They chose different models to interpret the observations, but the observations were empirical. Let us compare some of the old legends with modern scientific observations about the geological history of the Indian subcontinent. We will discuss three examples, and each could be seen as fiction or hard fact or some combination of both:

1) The geology of Kashmir (India) has been studied for more than 150 years now. As a result of these studies, it is now known that due to the rise of the Pir Panjal range around 4 million years ago, a vast lake formed, blocking the drainage from the Himalayas. Subsequently, the river Jhelum emerged as a result of the opening of a fault near Baramula, draining out the lake about 85,000 years ago. This is accepted as the geological history of the Kashmir valley.

Now let us compare this to the old legend: In Kashmir there is a very old tradition which describes a vast lake, called Satisara, in the valley in very ancient times. Kalhana, a poet chronicler, wrote his book Rajatarangini, or 'The River of Kings', in 1150 AD. In this book, he mentions an ancient lake (Satisara) giving a reference from a still earlier text, Nilamata Purana.

Aurel Stein (1961), who translated Rajatarangini, describes the legend of Satisara in these words: "This legend is mentioned by Kalhana in the Introduction of his Chronicle and is related at great length inNilamata… The demon Jalodbhava who resided in this lake was invisible in his own element and refused to come out of the lake. Vishnu thereupon called upon his brother Balabhadra to drain the lake…" Ignoring the mythical struggles between gods and demons, the legend does depict an account resembling the draining out of the primeval lake.

2) The sea level on the West Coast of India, as elsewhere during the Ice Ages, was about 100 meters lower than today and started rising only after 16,000 years ago. This is the accepted eustatic history.

The related legend says that when Parasurama donated all his land to the Brahmins, the latter asked him how he could live on the land that he had already donated away. Parasurama went to the cliff on the seashore and threw his Parasu (hatchet) into the sea and the sea receded, and then he occupied the land that thus emerged. This is possibly a reference to the regression of the sea and the newly emerged land.

3) The third example is of the river Satluj, a tributary of the Indus today. In finding its new course, the Sarasvati braided into several channels. This is the accepted geology.

The relevant legend says that the holy sage Vashista wanted to commit suicide by jumping into the Sarasvati, but the river wouldn't allow such a sage to drown himself, and broke up into hundreds of shallow channels, hence its ancient name Satadru. Unless the early author of such a legend observed the braiding process of the Satluj, he could not have imagined such a legend. This is another instance of legends coinciding with a modern geological observation.

Theorizing the possible role of myths, Scott says: "The complimentarity of the literal and the figurative help us to realize that the distinction between myth and science is not structural, but procedural… Myths in a broader, paradigmatic sense are condensed expressions of root metaphors that reflect the genius of particular knowledge traditions… Numerous studies have found that the “anthropomorphic” paradigms of egalitarian hunters and horticulturalists not only generate practical knowledge consistent with the insights of scientific ecology, but simultaneously cultivate an ethic of environmental responsibility that for western societies has proven elusive.”

The Israelis have been very successful in rediscovering many lost technologies relevant to their environment and culture by investigating their ancient myths and traditions. Through this, they have become pioneers in many processes of economic value that conventional European technology lacks.

The Goal

India’s intellectual resources are not limited to (though they are limited by) its ‘Indi-genius’ doubting intellectual elite. Today, there are Indian economists, social developers, and scholars who are working hard to revitalize many TKS. Resources for research and teaching of India’s Traditional Knowledge Systems should be made available for the following reasons:

India has amongst the best cases for successful revival of TKS: It has a rich heritage still intact in this area. It has the largest documented ancient literature relevant to TKS. It has the intellectual resources to appreciate this and to implement this revival, provided the Macaulayite mental blocks could be shaken up through re-education of its governing elite. It has dire needs to diversify beyond dependence solely upon the new panacea of globalization and westernization.

India’s scientific heritage, besides its philosophical and cultural legacy, needs to be properly understood. The aim is not inspired by chauvinism, but to understand the genius of Indian civilization better. This would overhaul the current assessment of India’s potential.

To correct the portrayal of the history of science, the history of ideas, mainstream accounts of world history, anthropology and culture. This entails emphasizing to scholars and educators that TKS should be included, especially India’s achievements and contributions to world science that have been very significant but unappreciated.

To include Traditional Knowledge Systems in economic planning, because they are eco-friendly, sustainable, labor rather than capital intensive, and more available to the masses. This should be done in parallel with the top down ‘modern’ scientific development using westernized ‘globalization’, as the two should co-exist and each should be used based on its merits.

TKS and Inner Sciences, History, and Society Today

Inner Sciences: The Inner Sciences of India have been on the one hand appropriated by the west, and on the hand have been depicted as being in conflict with the progressive, rational, and materialistic west. In fact, inner and outer realms are often viewed as opposites, that can at best be balanced because one contradicts the other. This assumes that Inner Sciences make a person and society less productive, creative, and competitive in the outer realm. However, India’s TKS are empirical evidence to demonstrate that Inner Sciences and outer development did co-exist in a mutually symbiotic relationship. This is a major reason to properly study India’s TKS.

Without removing this tension between inner and outer, it would be difficult to seriously motivate the modern world to advance in the Inner Sciences in a major way. Inner progress without the outer would be a world negating worldview, which India’s TKS record shows not to be the case in classical India. Outer progress without inner cultivation results in societies that are too materialistic, too selfish to the point of genocides and holocausts, eco-unfriendly, and dependent upon force and control for social harmony.

History: Until the 1800s, TKS generated large scale economic productivity for Indians. It was the TKS based thriving Indian economy that attracted so many waves of invaders, culminating with the British. Traditionally, India was one of the richest regions in the world, and most Indians were neither 'backward' nor uneducated nor poor. Some historians have recently begun to come out with this side of the story, demonstrating that it was massive economic drainage, oppression, social re-engineering, and so forth at the hands of colonizers that made millions of 'new poor' over the past few centuries. This explanation yields a radically different reading of the poverty in India today. Upon acknowledging India's traditional knowledge systems, one is forced to discard accounts of its history that essentialize its poverty and the accompanying social evils. The reality of TKS contradicts notions such as:

India was less rational and scientific than the west.

India was world negating in its outlook (which is a misreading of the Inner Sciences), and hence did not advance itself from within.

India's civilization was mainly imported via invaders, except for its problems such as caste that were its own 'essences'.

Indian society was socially backward (to the point of being seen as lacking in morality); hence it depends upon westernization to reform its current problems.

Society Today: Is India a ***'developing' society, or is it a 're-developing' society?*** Without appreciating the TKS of a people, how could anthropologists and sociologists interpret the current condition of a society? Were they always poor, always living in polluted and socially problematic conditions as today, in which case these problems are essences? Or is there a history behind the present condition? This history should not, however, excuse the failures of fifty years of independence to deal properly with the economic and social problems that persist.

Going forward, Traditional Knowledge Systems are eco-friendly, symbiotic with the environment, and therefore can help provide a sustainable lifestyle. Since the benefits of heavy industries do not trickle down to the people below the poverty line or to so-called developing countries, a revival of traditional technologies and crafts must complement the modern 'development' schemes for eradication of poverty. In this regard, the distinction between elite and folk science was non-existent in ancient times: India's advanced metallurgy and civil engineering was researched and practiced by artisan guilds.